Chapter 7 Body coordination

EXTRA WORKSHEET 1: Endocrine system

Hormones are chemical substances that act as messengers in the body. They are produced in endocrine glands scattered throughout the body. Together, all these glands form the endocrine system.

A part of the brain called the hypothalamus constantly checks the internal environment of the body and, if these conditions change, the hypothalamus sends signals to the nearby pituitary gland. The pituitary gland responds to these signals by either secreting hormones or releasing less of a particular hormone. The hormones from the pituitary gland control the activities of other endocrine glands such as the thyroid, parathyroid, thymus, adrenal glands, pancreas, ovaries and testes. The endocrine system controls important aspects of the body such as body temperature, rate of metabolism and water content.

Although hormones reach all your body cells they do not affect all the cells. Hormones are specific and act only on their target cells.

Questions

1 **Name** the parts of the endocrine system labelled A to G in Figure 1.

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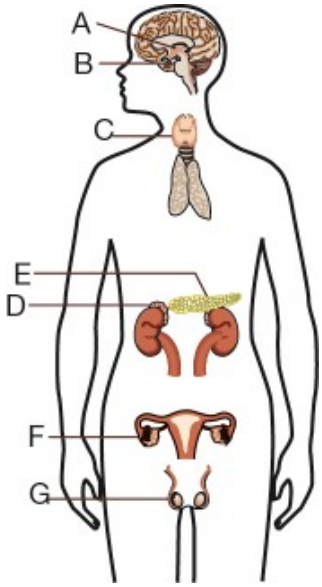


Figure 1 Endocrine system.

2 **Use** the following terms to complete the sentences below.

receptors **blood stream** **target cells** **endocrine glands** **hormones**

The chemical messengers of the endocrine system are known as _____. They are produced in _____. Hormones move in the _____ to reach every cell in the body. However, not every cell responds to the message. Only cells with specific _____ react. They are _____.

3 **Compare** the nervous and endocrine systems by completing the table.

Nervous system	Endocrine system
Messages travel along _____	Messages travel in the _____
Messages are in the form of _____	Messages are in the form of _____

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Messages travel in _____ direction.	Messages travel throughout the body but only act on _____
Messages cause very _____ reactions.	Messages cause _____ changes.
Controls things like: _____ _____	Controls things like: _____ _____

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